Table 1: **Calibration under Covariate and Concept shifts.** G-ΔUQ leads to better calibrated models for node-(GOODCora) and graph-level prediction tasks under different kinds of distribution shifts. Notably, G-ΔUQ can be combined with post-hoc calibration techniques to further improve calibration. The expected calibration error (ECE) is reported. Best, Second.

			Shift: Concept				Shift: Covariate			
			Accuracy (↑)		ECE (↓)		Accuracy (↑)		ECE (↓)	
Dataset	Domain	Calibration	No G-Δ UQ	G-Δ UQ	No G-Δ UQ	G-Δ UQ	No G-Δ UQ	G-Δ UQ	No G-Δ UQ	G-Δ UQ
GOODCora	Degree	×	0.581±0.003	0.595±0.003	0.307±0.009	0.13±0.011	0.47±0.002	0.518±0.014	0.348±0.032	0.141±0.008
		CAGCN	0.581±0.003	**0.597**±0.002	0.135±0.009	0.128±0.025	0.47±0.002	0.522±0.025	0.256±0.08	0.231±0.025
		Dirichlet	0.534±0.007	0.551±0.004	0.12±0.004	0.196±0.003	0.414±0.007	0.449±0.01	0.163±0.002	0.356±0.01
		ETS	0.581±0.003	0.596±0.004	0.301±0.009	0.116±0.018	0.47±0.002	**0.523**±0.003	0.31±0.077	0.141±0.003
		GATS	0.581±0.003	0.596±0.004	0.185±0.018	0.229±0.039	0.47±0.002	0.521±0.011	0.211±0.004	0.308±0.011
		IRM	0.582±0.002	0.597±0.002	0.125±0.001	0.102±0.002	0.469±0.001	0.522±0.004	0.194±0.005	0.13±0.004
		Orderinvariant	0.581±0.003	0.592±0.002	0.226±0.024	0.213±0.049	0.47±0.002	0.498±0.027	0.318±0.042	0.196±0.027
		Spline	0.571±0.003	0.595±0.003	0.080±0.004	**0.068**±0.004	0.459±0.003	0.52±0.004	0.158±0.01	**0.098**±0.004
		VS	0.581±0.003	0.596±0.004	0.306±0.004	0.127±0.002	0.47±0.001	0.522±0.005	0.345±0.005	0.146±0.005
GOODCMNIST	Color	×	0.499±0.003	0.497±0.002	0.439±0.078	0.334±0.066	0.348±0.009	0.355±0.034	0.551±0.147	0.423±0.172
		Dirichlet	0.495±0.009	0.510±0.008	0.303±0.012	0.304±0.007	0.350±0.053	0.335±0.059	0.542±0.091	0.406±0.076
		ETS	0.499±0.011	0.500±0.013	0.433±0.014	0.359±0.013	0.348±0.037	0.336±0.067	0.538±0.077	0.467±0.088
		IRM	0.499±0.006	0.500±0.010	0.285±0.004	**0.283**±0.008	0.348±0.049	0.336±0.071	0.416±0.084	0.425±0.093
		Orderinvariant	0.499±0.030	0.500±0.028	0.379±0.050	0.386±0.042	0.348±0.036	0.337±0.059	0.475±0.077	0.542±0.104
		Spline	0.495±0.008	0.497±0.010	0.29±0.007	0.291±0.008	0.346±0.051	0.335±0.071	0.414±0.085	0.425±0.093
		VS	0.499±0.007	0.500±0.012	0.439±0.006	0.377±0.009	0.349±0.037	0.336±0.067	0.549±0.071	0.468±0.089
		Ensembling	0.505±0.001	**0.509**±0.004	0.437±0.082	0.343±0.004	0.397±0.005	**0.408**±0.006	0.423±0.017	**0.327**±0.013
GOODMotif	Basis	×	0.925±0.001	0.925±0.003	0.095±0.014	0.078±0.007	0.691±0.001	0.689±0.002	0.329±0.274	0.342±0.266
		Dirichlet	0.925±0.011	0.923±0.010	0.081±0.015	0.103±0.007	0.686±0.009	0.681±0.009	0.337±0.067	0.316±0.047
		ETS	0.925±0.009	0.927±0.012	0.095±0.010	0.096±0.013	0.691±0.011	0.699±0.016	0.314±0.041	0.304±0.049
		IRM	0.925±0.014	0.93±0.013	0.087±0.018	0.097±0.010	0.691±0.011	0.698±0.016	0.316±0.051	0.305±0.045
		Orderinvariant	0.925±0.010	0.928±0.011	0.091±0.009	0.093±0.007	0.691±0.011	0.690±0.011	0.321±0.050	0.319±0.041
		Spline	0.925±0.010	0.927±0.011	0.091±0.008	0.089±0.012	0.691±0.010	0.689±0.016	0.324±0.055	0.313±0.051
		VS	0.925±0.009	0.927±0.012	0.095±0.010	0.095±0.013	0.683±0.013	0.680±0.018	0.326±0.057	0.311±0.059
		Ensembling	0.932±0.002	**0.943**±0.006	0.086±0.016	**0.047**±0.003	**0.714**±0.012	0.699±0.009	**0.298**±0.383	0.321±0.196
GOODSST2	Length	×	0.694±0.002	0.693±0.001	0.288±0.017	0.277±0.011	0.826±0.002	0.828±0.004	0.159±0.027	0.154±0.039
		Dirichlet	0.686±0.02	0.683±0.001	0.15±0.021	**0.138**±0.015	0.793±0.005	0.8±0.012	0.15±0.02	**0.131**±0.007
		ETS	0.685±0.02	0.683±0.001	0.21±0.009	0.211±0.003	0.794±0.005	0.8±0.011	0.287±0.007	0.296±0.014
		IRM	0.685±0.019	0.682±0.002	0.239±0.002	0.231±0.006	0.796±0.006	0.801±0.011	0.26±0.005	0.265±0.011
		Orderinvariant	0.685±0.02	0.683±0.001	0.225±0.002	0.222±0.003	0.794±0.005	0.8±0.011	0.226±0.003	0.224±0.007
		Spline	0.684±0.02	0.683±0.002	0.233±0.005	0.23±0.005	0.79±0.004	0.794±0.016	0.259±0.005	0.263±0.012
		VS	0.685±0.019	0.683±0	0.334±0.044	0.374±0.002	0.787±0.008	0.8±0.013	0.307±0.116	0.32±0.011
		Ensembling	0.705±0.002	**0.709**±0.004	0.276±0.038	0.248±0.022	0.838±0.001	**0.842**±0.006	0.154±0.032	0.132±0.019

Experimental Setup. We use three different datasets (GOODCMNIST, GOODMotif-basis, GOODSST2) with their corresponding splits and shifts from the recently proposed Graph Out-Of Distribution (GOOD) benchmark (Gui et al., 2022). The architectures and hyperparameters suggested by the benchmark are used for training. G-ΔUQ uses `READOUT` anchoring and 10 random anchors (see App. A.7 for more details). We report accuracy and expected calibration error for the OOD test dataset, taken over three seeds.

Results. As shown in Table 1, we observe that G-ΔUQ leads to inherently better calibrated models, as the ECE from G-ΔUQ without additional post-hoc calibration (×) is better than the vanilla ("No G-ΔUQ") counterparts on 5/6 datasets. Moreover, we find that combining G-ΔUQ with a particular post-hoc calibration methods further elevates its performance relative to combining the same strategy with vanilla models. Indeed, for a fixed post-hoc calibration strategy, G-ΔUQ improves the calibration, while maintaining comparable if not better accuracy on the vast majority of the methods and datasets. There are some settings where combining G-ΔUQ or the vanilla model with a post-hoc method leads decreases performance (for example, GOODSST2, covariate, ETS, calibration) but we emphasize that this is not a short-coming of G-ΔUQ. Posthoc strategies, which rely upon ID calibration datasets, may not be effective on shifted data. This further emphasizes the importance of our OOD evaluation and G-ΔUQ as an intrinsic method for improving uncertainty estimation.

5.3 Using Confidence Estimates in Safety-Critical Tasks

While post-hoc calibration strategies rely upon an additional calibration dataset to provide meaningful uncertainty estimates, such calibration datasets are not always available and may not necessarily improve OOD performance (Ovadia et al., 2019). Thus, we also evaluate the quality of the uncertainty estimates directly provided by G-ΔUQ on two additional UQ-based, safety-critical tasks (Hendrycks et al., 2022b; 2021; Trivedi et al., 2023b): (i) OOD detection (Hendrycks et al., 2019), which attempts to classify samples as in- or out-of-distribution, and (ii) generalization error prediction (GEP) (Jiang

Table 2: **GOOD-Datasets, OOD Detection Performance.** The AUROC of the binary classification task of classifying OOD samples is reported. G-ΔUQ variants outperform the vanilla models on 6/8 datasets. [We further note that end-to-end G-ΔUQ does in fact lose performance relative to the vanilla model on 4 datasets. Investigating why pretrained G-ΔUQ is able to increase performance on those datasets is an interesting direction of future work. It does not appear that a particular shift is more difficult for this task: concept shift is easier for GOODCMNIST and GOODMotif(Basis) while covariate shift is easier for GOODMotif(Size) and GOODSST2. Combining G-ΔUQ with more sophisticated, uncertainty or confidence based OOD scores may further improve performance.]

	CMNIST (Color)		MotifLPE (Basis)		MotifLPE (Size)		SST2	
Method	**Concept(↑)**	**Covariate(↑)**	**Concept(↑)**	**Covariate(↑)**	**Concept(↑)**	**Covariate(↑)**	**Concept(↑)**	**Covariate(↑)**
Vanilla	0.759 ± 0.006	0.468 ± 0.092	0.736 ± 0.021	$\mathbf{0.466 \pm 0.001}$	0.680 ± 0.003	0.755 ± 0.074	$\mathbf{0.350 \pm 0.014}$	0.345 ± 0.066
G-ΔUQ	0.771 ± 0.002	0.470 ± 0.043	0.758 ± 0.006	0.328 ± 0.022	0.677 ± 0.005	0.691 ± 0.067	0.338 ± 0.023	0.351 ± 0.042
Pretr. G-ΔUQ	$\mathbf{0.774 \pm 0.016}$	$\mathbf{0.543 \pm 0.152}$	$\mathbf{0.769 \pm 0.029}$	0.272 ± 0.025	$\mathbf{0.686 \pm 0.004}$	$\mathbf{0.829 \pm 0.113}$	0.324 ± 0.055	$\mathbf{0.446 \pm 0.049}$

Table 3: **RotMNIST-Calibration.** Here, we report expanded results (calibration) on the Rotated MNIST dataset, including a variant that combines G-ΔUQ with Deep Ens. Notably, we see that anchored ensembles outperform basic ensembles in both accuracy and calibration.

Architecture	LPE?	G-ΔUQ	Calibration	Avg.ECE (↓)	ECE (10) (↓)	ECE (15) (↓)	ECE (25) (↓)	ECE (35) (↓)	ECE (40) (↓)
GatedGCN	×	×	×	0.038 ±0.001	0.059 ±0.001	0.068 ±0.340	0.126 ±0.008	0.195 ±0.012	0.245 ±0.011
	×	✓	×	**0.018** ±0.008	<u>0.029</u> ±0.013	**0.033** ±0.164	<u>0.069</u> ±0.033	<u>0.117</u> ±0.048	<u>0.162</u> ±0.067
	×	×	Ensembling	0.026 ±0.000	0.038 ±0.001	0.042 ±0.001	0.084 ±0.002	0.135 ±0.001	0.185 ±0.003
	×	✓	Ensembling	**0.014** ±0.003	**0.018** ±0.005	**0.021** ±0.005	<u>0.036</u> ±0.012	<u>0.069</u> ±0.032	<u>0.114</u> ±0.056
GatedGCN	✓	×	×	0.036 ±0.003	0.059 ±0.002	0.068 ±0.340	0.125 ±0.006	0.191 ±0.007	0.240 ±0.008
	✓	✓	×	0.022 ±0.007	**0.028** ±0.014	<u>0.034</u> ±0.169	**0.062** ±0.022	**0.109** ±0.019	**0.141** ±0.019
	✓	×	Ensembling	0.024 ±0.001	0.038 ±0.001	0.043 ±0.002	0.083 ±0.001	0.139 ±0.004	0.181 ±0.002
	✓	✓	Ensembling	0.017 ±0.002	0.024 ±0.005	<u>0.027</u> ±0.008	**0.030** ±0.004	**0.036** ±0.012	**0.059** ±0.033
GPS	✓	×	×	0.026 ±0.001	0.044 ±0.001	0.052 ±0.156	0.108 ±0.006	0.197 ±0.012	0.273 ±0.008
	✓	✓	×	0.022 ±0.001	0.037 ±0.005	0.044 ±0.133	0.091 ±0.008	0.165 ±0.018	0.239 ±0.018
	✓	×	Ensembling	<u>0.016</u> ±0.001	0.026 ±0.002	0.030 ±0.000	0.066 ±0.000	0.123 ±0.000	0.195 ±0.000
	✓	✓	Ensembling	**0.014** ±0.000	<u>0.023</u> ±0.002	<u>0.027</u> ±0.003	0.055 ±0.004	0.103 ±0.006	0.164 ±0.006

et al., 2019), which attempts to predict the generalization on unlabeled test datasets (to the best of our knowledge, we are the first to study GEP of graph classifiers). In the interest of space, we present the results on GEP in the appendix.

OOD Detection Experimental Setup. By reliably detecting OOD samples and abstaining from making predictions on them, models can avoid over-extrapolating to irrelevant distributions. While many scores have been proposed for detection (Hendrycks et al., 2019; 2022a; Lee et al., 2018; Wang et al., 2022a; Liu et al., 2020), popular scores, such as maximum softmax probability and predictive entropy (Hendrycks & Gimpel, 2017), are derived from uncertainty estimates. Here, we report the AUROC for the binary classification task of detecting OOD samples using the maximum softmax probability as the score (Kirchheim et al., 2022).

OOD Detection Results. As shown in Table 2, we observe that G-ΔUQ variants improve OOD detection performance over the vanilla baseline on 6/8 datasets, where pretrained G-ΔUQ obtains the best overall performance on 6/8 datasets. G-ΔUQ performs comparably on GOODSST2(concept shift), but does lose some performance on GOODMotif(Covariate). We note that vanilla models provided by the original benchmark generalized poorly on this particular dataset (increased training time/accuracy did not improve performance), and this behavior was reflected in our experiments. We suspect that poor generalization coupled with stochasticity may explain G-ΔUQ's performance here.

6 FINE GRAINED ANALYSIS OF G-ΔUQ

Given that the previous sections extensively verified the effectiveness of G-ΔUQ on a variety of covariate and concept shifts across several tasks, we seek a more fine-grained understanding of G-ΔUQ's behavior with respect to different architectures and training strategies. In particular, we demonstrate that G-ΔUQ continues to improve calibration with expressive graph transformer architectures, and that using `READOUT` anchoring with pretrained GNNs is an effective lightweight strategy for improving calibration of frozen GNN models.

6.1 Calibration under Controlled Shifts

Recently, it was shown that modern, non-convolutional architectures (Minderer et al., 2021) are not only more performant but also more calibrated than older, convolutional architectures (Guo et al., 2017) under vision distribution shifts. Here, we study an analogous question: are more expressive GNN architectures better calibrated under distribution shift, and how does G-ΔUQ impact their calibration? Surprisingly, we find that more expressive architectures are not considerably better calibrated than their MPNN counterparts, and ensembles of MPNNs outperform ensembles of GTrans. Notably, G-ΔUQ continues to improve calibration with respect to these architectures as well.

Experimental Setup. *(1) Models.* While improving the expressivity of GNNs is an active area of research, positional encodings (PEs) and graph-transformer (GTran) architectures (Müller et al., 2023) are popular strategies due to their effectiveness and flexibility. GTrans not only help mitigate over-smoothing and over-squashing (Alon & Yahav, 2021; Topping et al., 2022) but they also better capture long-range dependencies (Dwivedi et al., 2022b). Meanwhile, graph PEs help improve expressivity by differentiating isomorphic nodes, and capturing structural vs. proximity information (Dwivedi et al., 2022a). Here, we ask if these enhancements translate to improved calibration under distribution shift by comparing architectures with/without PEs and transformer vs. MPNN models. We use equivariant and stable PEs (Wang et al., 2022b), the state-of-the-art, "general, powerful, scalable" (GPS) framework with a GatedGCN backbone for the GTran, GatedGCN for the vanilla MPNN, and perform READOUT anchoring with 10 random anchors. *(2) Data.* In order to understand calibration behavior as distribution shifts become progressively more severe, we create structurally distorted but valid graphs by rotating MNIST images by a fixed number of degrees (Ding et al., 2021) and then creating the corresponding superpixel graphs (Dwivedi et al., 2020; Knyazev et al., 2019; Velickovic et al., 2018). (See Appendix, Fig. 6.) Since superpixel segmentation on these rotated images will yield different superpixel k-nn graphs but leave class information unharmed, we can emulate different severities of label-preserving structural distortion shifts. We note that models are trained only using the original (0° rotation) graphs. Accuracy (see appendix) and ECE over 3 seeds are reported for the rotated graphs.

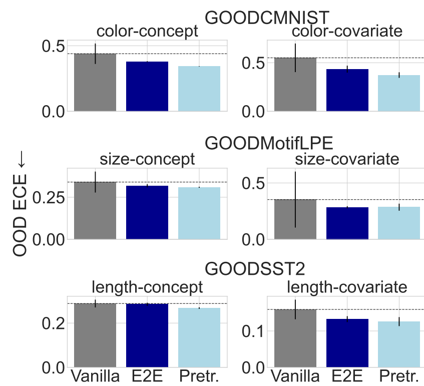

Figure 4: **G-ΔUQ with Pretrained Models.** When used with pretrained models, G-ΔUQ achieves strong out-of-distribution calibration performance with respect to both vanilla models and those models trained with end-to-end G-ΔUQ.

Results. In Table 3, we present the OOD calibration results, with results of more variants and metrics in the supplementary Table 5 and 8. First, we observe that PEs have minimal effects on both calibration and accuracy by comparing GatedGCN with and without LPEs. This suggests that while PEs may enhance expressivity, they do not directly induce better calibration. Next, we find that while vanilla GPS is better calibrated when the distribution shift is not severe (10, 15, 25 degrees), it is less calibrated (but more performant) than GatedGCN at more severe distribution shifts (35, 40 degrees). This is in contrast to known findings about vision transformers. Lastly, we see that G-ΔUQ continues to improve calibration across all considered architectural variants, with minimal accuracy loss. Surprisingly, however, we observe that *ensembles* of G-ΔUQ models not only effectively resolve any performance drops, they also cause MPNNs to be better calibrated than their GTran counterparts.

6.2 How does G-ΔUQ perform with pretrained models?

As large-scale pretrained models become the norm, it is beneficial to be able to perform lightweight training that leads to safer models. Thus, we investigate if READOUT anchoring is such a viable strategy when working with pretrained GNN backbones, as it only requires training a stochastically centered classifier on top of a frozen backbone. (Below, we discuss results on GOODDataset, but please see A.4 for results on RotMNIST and A.12 for additional discussion.)